

PUQI ZHOU

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RESEARCH INTERESTS

My research interests **Human-Computer Interaction**, **Human-Robot Interaction**, **Robotics**, *Computer Vision*, and *Mixed Reality* to support domain experts in high-stakes, time-critical environments through multi-robot systems around **human attention**. This is **important** because multi-robot systems have the potential to transform public safety, search & rescue, and other high-stakes domains. However, this is **difficult** because humans and robots often operate with different situational awareness and contextual understanding, making seamless task communication and coordination a persistent challenge.

EDUCATION

Ph.D.	George Mason University (GMU), Fairfax, VA, USA Computer Science Advisor: David Porfirio & Sungsoo Ray Hong	2021–present
B.Eng.	University College Dublin (UCD), Dublin, Ireland Beijing University of Technology (BJUT), Beijing, China Dual Degree in Electronic and Information Engineering Merit Student (Top 5%), Outstanding Student Cadre (Top 5%)	2017–2021

PUBLICATIONS

- Conferences
- [C5] **Designing Multi-Robot Ground Video Sensemaking with Public Safety Professionals**
PUQI ZHOU, Ali Asgarov, Aafiya Hussain, Wonjoon Park, Amit Paudyal, Sameep Shrestha, Chia-Wei Tang, Michael F. Lighthiser, Michael R. Hieb, Xuesu Xiao, Chris Thomas, and Sungsoo Ray Hong
Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems. [CHI26] [Project] [Paper]
- [C4] **Theia: Gaze-driven and Perception-aware Volumetric Content Delivery for Mixed Reality Headsets**
Nan Wu, Kaiyan Liu, Ruizhi Cheng, Bo Han, and **PUQI ZHOU**
Proceedings of the 22nd ACM Conference on Mobile Systems, Applications and Services. [MobiSys24] [Paper]
- [C3] **Muv2: Scaling up Multi-user Mobile Volumetric Video Streaming via Content Hybridization and Sharing**
Yu Liu, **PUQI ZHOU**, Zejun Zhang, Anlan Zhang, Bo Han, Zhenhua Li, and Feng Qian
Proceedings of the 30th ACM Conference on Mobile Computing and Networking. [MobiCom24] [Paper]
- [C2] **Collaborative Online Learning with VR Video: Roles of Collaborative Tools and Shared Video Control**
Qiao Jin, Yu Liu, Ruixuan Sun, Chen Chen, **PUQI ZHOU**, Bo Han, Feng Qian, and Svetlana Yarosh
Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems. [CHI23] [Paper]
- [C1] **M5: Facilitating Multi-user Volumetric Content Delivery with Multi-lobe Multicast over mmWave**
Ding Zhang, **PUQI ZHOU**, Bo Han, and Parth H. Pathak
Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems. [SenSys22] [Paper]
- Others

- [O1] **Volumivive: An Authoring System for Adding Interactivity to Volumetric Video**
Qiao Jin, Yu Liu, **PUQI ZHOU**, Bo Han, Svetlana Yarosh, and Feng Qian
2023 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops. [VR23]

RESEARCH EXPERIENCE

George Mason University (GMU)

Ph.D. Student

09/2025–Present

Department of Computer Science

Advisor: Prof. David Porfirio

Project: Robot Adaptive Task Planning around Human Attention

George Mason University (GMU)

Ph.D. Student

08/2023–Present

Department of Computer Science

Advisors: Prof. Sungsoo (Ray) Hong, Prof. Chris Thomas, Prof. Xuesu Xiao, and Prof. Michael Hieb

Project: Multi-Ground Robots Video Sensemaking for Public Safety

University College Dublin (UCD)

Research Assistant

09/2020–07/2021

Advisor: Prof. Xingqi Zhang

Project: Indoor Wireless Positioning

HONORS AND AWARDS

George Mason University Doctoral Research Scholars

2026

2026-2027, *Tire 1*

TEACHING EXPERIENCE

Teaching Assistant, GMU

CS 551, Computer Graphics.

Fall 2023, Spring 2024, Fall 2025

CS 695, Immersive Computing.

Spring 2023

CS 600, Theory of Computation.

Spring 2026

MENTORSHIP

Darsh Shetty (2025)—*Academy of Engineering and Technology*

Wonjoon Park (2024-present)—*UMD*

Karthik Nutulapati (2024-2025)—*GMU*

ACADEMIC SERVICE

Referee Service

CHI - ACM Conference on Human Factors in Computing Systems

2023, 2025, 2026

IUI - ACM Conference on Intelligent User Interfaces

2026

CSCW - ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing

2025

Event Organization

ACM SenSys N2Women (Networking Women)

2022

TECHNICAL SKILLS

Programming Languages

Python, Java, JavaScript, HTML, CSS, C#, C, PDDL

Robotics Tools, Libraries, Frameworks and Platforms

ROS2, Mujoco Simulator, OpenCV, Unity Game Engine, Hello-Robot Stretch 3, Frodo Robot

Design

Figma, Affinity